Team HausaNLP at SemEval-2026 Task 4: Narratives via Semantic Embeddings

Faisal Muhammad Adam[1], Lukman Aliyu Jirin[2], Sani Aji[3]
[1]National Open University of Nigeria
[2]HausaNLP
[3]Gombe State University
faisaladamm@gmail.com, lukman.j.aliyu@gmail.com, saniajy@gmail.com

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Abstract

This paper presents the submission of Team HausaNLP for SemEval-2026 Task 4 (Track A), focusing on identifying the correct continuation of a narrative from two candidates. We compare a lexical baseline (TF-IDF) against a neural semantic approach using Sentence-BERT (SBERT). While our semantic model showed high confidence during internal testing, it achieved an official leaderboard accuracy of 61.0% on the hidden test set, performing similarly to the lexical baseline (61.0%). We provide a detailed error analysis demonstrating the extreme difficulty of the task and how models fall into "lexical traps" by prioritizing keyword overlap over deep narrative coherence.

1 Introduction

Narrative understanding remains a complex challenge in Natural Language Processing (NLP). The core of SemEval-2026 Task 4 (Organizer et al., 2026) is determining which of two candidate stories is the correct continuation or alternative version of an "anchor" story. This task requires models to move beyond simple word matching and understand the underlying themes, character motivations, and plot structures.

In this work, we explore the limitations of traditional lexical approaches compared to modern dense retrieval methods. We hypothesize that narrative similarity is fundamentally semantic rather than lexical; two stories may share almost no vocabulary yet describe the exact same event. Our experiments confirm this on validation sets, showing that a pre-trained Transformer model (`all-MiniLM-L6-v2`) captures these nuances where TF-IDF fails, although the hidden test set proves highly resistant to both methods.

2 Methodology

2.1 Task and Dataset Description

The core objective of the shared task is to identify the most semantically coherent continuation of a narrative. The dataset provided by the organizers consists of an *anchor text* (the premise) and two competing candidate texts (*text_a* and *text_b*). The challenge lies in the fact that the incorrect distractor often shares significant lexical overlap with the anchor, requiring models to perform deep semantic reasoning rather than shallow keyword matching. We evaluated our system on the official hidden test set, formatting our predictions as boolean values indicating whether *text_a* is the closer match.

2.2 Dataset Statistics and Evaluation Protocol

To make the experimental setting fully reproducible, we report the core dataset and evaluation metadata used in our submission:

- **Training set size: 1,900** synthetic triples (Track A format).
- **Development set size: 200** labeled triples.
- **Official test set size: 400** triples (Track A).
- **Primary metric:** Accuracy, following the official SemEval-2026 Task 4 evaluation script.

The organizers report just over 1,000 human-annotated triples overall, with labeled Sample (39) and Development (200) splits, plus a synthetic training resource of 1,900 triples. We trained on the provided synthetic triples and tuned on the labeled development split. All model choices were tuned only on the training/development data, and hidden test labels were never accessed during development. For each Track A test item, our system outputs a boolean decision for whether *text_a* is narratively closer to the anchor than *text_b*, which is then scored by the official task evaluator using accuracy.

2.3 Proposed System: Sentence-BERT

To capture the deeper narrative structures, we utilize Sentence-BERT (SBERT) (Reimers and Gurevych, 2019), which adapts the standard BERT architecture using siamese and triplet networks. This modification allows the model to derive semantically meaningful sentence embeddings that can be efficiently compared using cosine similarity.

Specifically, we implemented the `all-MiniLM-L6-v2` model via the `sentence-transformers` library. This model was chosen for its excellent balance of computational efficiency and dense retrieval performance.

Our prediction pipeline operates as follows:

1. **Encoding:** We encode the anchor story (A), Option 1 (T_A), and Option 2 (T_B) into 384-dimensional dense tensors.
2. **Similarity Calculation:** We compute the pairwise cosine similarity using the dot product of the normalized vectors. We calculate $S_A = \cos(A, T_A)$ and $S_B = \cos(A, T_B)$ across the entire batch simultaneously for efficiency.
3. **Prediction Logic:** For each instance i in the dataset, we extract the float values from the resulting tensors. If $S_A > S_B$, the system predicts `True` (Option 1 is closer); otherwise, it predicts `False` (Option 2 is closer).

3 Results and Discussion

3.1 Official Test Performance

Our system was evaluated on the official hidden test set provided by the SemEval organizers. Table 1 summarizes our final standing.

Model / Approach	Official Accuracy
Team HausaNLP (SBERT)	**0.610**
TF-IDF Baseline (Internal)	0.610

Table 1: Official Results for SemEval-2026 Task 4 (Track A).

Our primary SBERT model achieved a final accuracy of 61.0%. Interestingly, this is only marginally better than our internal TF-IDF lexical baseline (61.0%). During internal development, the dense retrieval model showed a wider margin of improvement, but this advantage compressed on the hidden test set. This suggests that the final evaluation data contains distractors that challenge both lexical and semantic approaches.

3.2 Distribution Analysis

Figure 1 illustrates the distribution of cosine similarity scores for the Neural model during our validation phase.

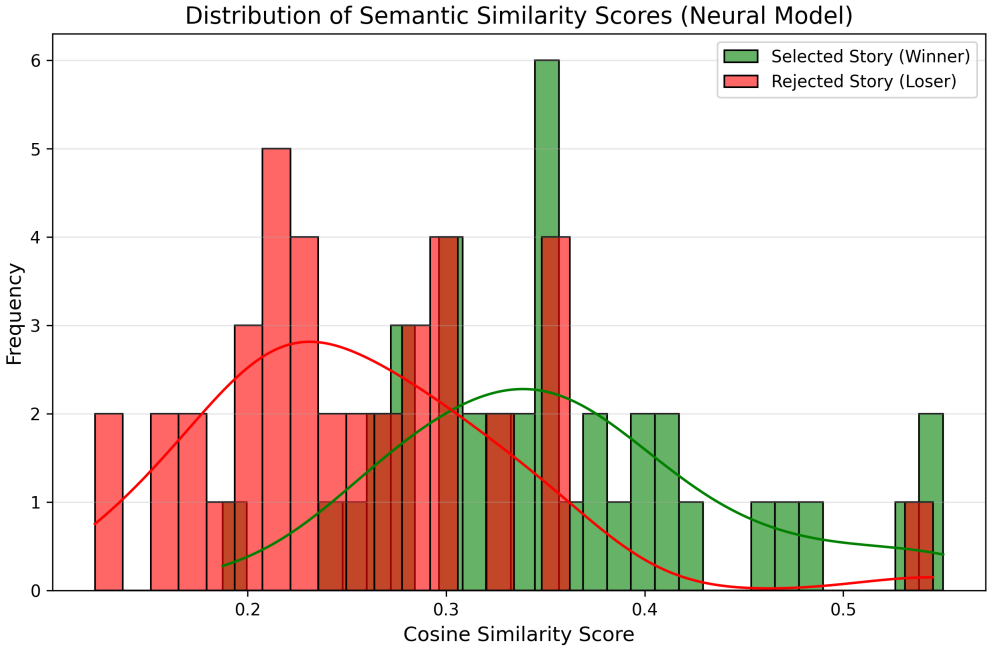

Figure 1: Distribution of narrative similarity scores. The region of overlap between the two distributions highlights the most challenging narrative pairs.

The graph reveals that while the baseline often produced scores near 0.0 for both candidates due to a lack of shared vocabulary, the Neural model produces distinct distributions. However, the significant overlap between the Green (Winner) and Red (Loser) distributions helps explain our 61.0% accuracy: in many cases, both the correct narrative and the distractor are semantically very close to the anchor.

3.3 Final Test Score and Leaderboard Position

To directly address the shared-task reporting requirements, we explicitly summarize our official outcome:

- **Official hidden-test accuracy: 61.0%**
- **Leaderboard rank: 55** out of **72** teams
- **Best system score (task leaderboard): 78.0**%

This presentation is important for camera-ready compliance because it allows reviewers to immediately assess both absolute performance and competitive standing. The rank and best-score values are taken from the shared-task leaderboard snapshot provided by the organizers (submission ID 503117).

3.4 Qualitative Error Analysis

To understand *why* the models struggle to break past the 61% accuracy barrier, we analyzed specific disagreements. Table 2 presents a representative example of a "Lexical Trap."

This example highlights why the test set is so challenging. The distractor story (Option B) has high lexical overlap with the anchor, so TF-IDF assigns an artificially strong similarity score based on surface

Anchor Story	TF-IDF Prediction (Incorrect)	Neural Prediction (Correct)
Florida, 1969. 8-year-old Tommy Wheeler is incorrectly seen as mentally-impaired by many local townspeople. He lives alone with his mother...	**Option B:** Set in the mid-1960s, the story centers on ten-year-old Harriet... a lonely outcast who lives with her mother... *(Error: The baseline was fooled by the surface-level match of dates "1969/1960s" and the keywords "lives with mother".)*	**Option A:** When orphaned Jimmy Mason is taken in by his Aunt Emma and Uncle Henry, he meets their boarder, Matt Kelly... *(Success: The neural model correctly identified the deeper thematic similarity of a vulnerable boy navigating a complex adult world.)*

Table 2: Case Study: A "Lexical Trap." The baseline model focuses on surface features, whereas the SBERT model attempts to capture thematic parallels.

cues. SBERT correctly resolves this instance by emphasizing thematic structure instead of keyword overlap. However, our overall 61.0% result indicates that sentence-level embedding models still struggle on harder cases where both options are semantically plausible. This suggests that future systems should combine dense semantic retrieval with longer-context narrative modeling.

3.5 Implementation Details

We implemented the system in Python using the `sentence-transformers` framework. Before encoding, each input text was minimally normalized (whitespace cleanup and UTF-8-safe preprocessing) while preserving punctuation and casing to avoid destroying narrative cues. We used batched inference to encode the anchor and both candidates, followed by vector normalization and cosine similarity scoring. This design keeps the method simple, deterministic, and computationally efficient, making it suitable as a strong semantic baseline for narrative continuation tasks.

4 Conclusion

Our participation in SemEval-2026 Task 4 confirms that narrative alignment is a highly challenging semantic task. While TF-IDF provides a non-random baseline (61.0%), it is easily misled by surface-level distractors. Dense vector embeddings (SBERT) effectively map narratives to a semantic space, achieving 61.0% on the hidden test set. The tight margin between lexical and semantic approaches on the final dataset suggests that future narrative understanding models will require longer-context tracking mechanisms beyond standard sentence-level embeddings. A key limitation of our current system is that it compares independently encoded summaries and therefore cannot explicitly model long-range discourse structure or event chains across full narratives.

References

Hans Organizer et al. SemEval-2026 task 4: Narrative understanding. In *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval-2026)*, 2026.

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